Contents

% The ngspice `pyplot` command — feature package % ngspice-46 + OpenVAF-r enhancement project

What the feature is

`pyplot` is an ngspice interactive/`.control` command that plots simulation vectors with matplotlib — the same role `gnuplot` plays, but rendering through Python/matplotlib. It writes a self-contained `<name>.data` table and a `<name>.py` matplotlib script and then runs Python on the script, either opening an interactive window or, with a file terminal, rendering a headless image (PNG/SVG/PDF).

```
pyplot [file] <vectors|expressions> [title '...'] [xlabel ...] [ylabel ...]
       [xlimit lo hi] [ylimit lo hi]
```

The output file name is optional; a bare net name plots its node voltage (no `v()` needed), and every plot argument may be an expression (`v(out)`, `db(v(out))`, `out1+6`, …).

Options (**`set`** variables)

set variable	effect
`pyplot_terminal=png|svg|pdf`	render headless to `<name>.<fmt>` instead of a window
`pyplot_subplots=N`	N traces per stacked subplot (shared x-axis); 0/unset = one axis
`pyplot_style=<name>`	a matplotlib style sheet (`dark`, `ggplot`, `bmh`, …)
`pyplot_figsize=W,H`	figure size in inches
`pyplot_python=<exe>`	the Python interpreter (default `python3`)
`pyplot_linewidth=<w>`	line width in points for every trace *(Enhancement-183)*
`pyplot_backend=<name>`	matplotlib backend, e.g. `TkAgg`/`QtAgg`/`MacOSX`/`Agg` *(Enhancement-183)*
`pointstyle=markers`	draw points instead of lines

Example:

```
.control
run
set pyplot_terminal=png
set pyplot_linewidth=2.0
set pyplot_subplots=1
pyplot chain Vin out1 out2 out3 title 'inverter chain'
.endc
```

Files changed

Baseline = the pristine `original/ngspice-46` tree; the feature lives entirely in the `pyplot` command front-end. `windisp.c` also matches a grep for "pyplot" but its 16 mentions are pre-existing in the baseline (an unrelated Windows-display symbol) — it is *not* part of this feature and is not shipped.

File	Kind	Lines	Role
`src/frontend/com_pyplot.c`	new	94	the `pyplot` command: file-name/dir handling, dispatch
`src/frontend/com_pyplot.h`	new	5	its prototype
`src/frontend/plotting/pyplot.c`	new	329	`ft_pyplot()` — writes the `.data` table + `.py` script and runs Python
`src/frontend/plotting/pyplot.h`	new	15	`ft_pyplot()` prototype
`src/frontend/commands.c`	modified	—	registers `pyplot` in the command tables (`spcp_coms`, `nutcp_coms`)

File	Kind	Lines	Role
`src/frontend/plotting/plotit.c`	modified	—	routes `devname == "pyplot"` to `ft_pyplot()`, passing user x/y limits only when given
`src/frontend/Makefile.am`	modified	—	adds `com_pyplot.{c,h}` to the build
`src/frontend/plotting/Makefile.am`	modified	—	adds `pyplot.{c,h}` to the build

`Makefile.in` is generated from `Makefile.am` by `autogen.sh`/autoconf — only the `.am` files are shipped; regenerate the `.in` with the project's bootstrap.

The feature's lines in the shared (modified) files

`commands.c` — command registration (two dispatch tables):

```
#include "com_pyplot.h"
...
{ "pyplot", com_pyplot, FALSE, TRUE,
  "pyplot [file] plotargs : plot vectors with matplotlib" },
```

`plotit.c` — the dispatch to the pyplot back-end (only pins the axes for explicit user limits, otherwise leaves matplotlib to autoscale):

```
#include "pyplot.h"
...
if (devname && eq(devname, "pyplot")) {
    ft_pyplot(user_xlim ? xlims : NULL, user_ylim ? ylims : NULL,
              xdelta, ydelta, filename, title, xlabel, ylabel,
              gridtype, plottype, vecs);
    ...
}
```

Enhancement history

`pyplot` was built incrementally; each enhancement is documented in `enhancements_doc/`:

Enhancement	What it added
E-94	the `pyplot` command + PNG output (matplotlib back-end)
E-95	optional output file name (defaults to `pyplot`)
E-98	`pyplot_subplots`, `pyplot_style`, marker/point styles
E-99	SVG/PDF terminals + `pyplot_figsize`
E-182	autoscale by default (`fig.tight_layout()`); pin axes only for explicit `xlimit`/`ylimit`
E-183	distinct default names for successive no-name plots; write `.py`/`.data`/`.png` next to the `.cir`; `pyplot_linewidth`; `pyplot_backend`

Enhancement-183 (this package's newest changes)

1. Distinct default names — in window mode the Python viewer is launched in the background, so two plots that both omit the file name raced on the same `pyplot.py`/`pyplot.data` and both windows showed the *second* plot. Later no-name plots are now `pyplot-2`, `pyplot-3`, … (first stays `pyplot`).
2. Deck-folder output — the `.py`/`.data`/`.png` are written next to the circuit file (via `ngdirname(ci_filename)`), not ngspice's cwd; the script's `loadtxt`/`savefig` carry the full path. A bare in-folder deck name is unchanged (cwd).
3. **`pyplot_linewidth=<w>`** — line width for every trace.
4. **`pyplot_backend=<name>`** — explicit matplotlib backend (`matplotlib.use`), overriding the automatic choice.

How it works

1. `com_pyplot()` peels an optional leading file name (a word that is not itself a plot expression / vector), defaults it to `pyplot` (uniquified per session), and — for a bare name — prefixes the circuit file's directory. It then calls the shared `plotit()` with device `"pyplot"`.
2. `plotit()` parses the vector expressions and, recognizing the `pyplot` device, calls `ft_pyplot()`.

3. `ft_pyplot()` writes the numeric data to `<name>.data`, emits a matplotlib script `<name>.py` (backend/style/figsize/subplots/linewidth honored, title and labels quoted, axes autoscaled unless the user gave limits), and runs `python3  <name>.py` — synchronously for a file terminal, in the background for an interactive window.

Verification

`examples/pyplot_examples/verify_pyplot.py` — 13 checks under both ngspice linear solvers: the OSDI-model transient renders a valid PNG whose script plots the requested vectors; autoscale vs explicit-limit behavior; an AC log-scale plot; the optional-file-name default; and the four Enhancement-183 behaviors (distinct default names with correct titles, deck-folder output, `linewidth`, `backend`).

Archive contents

```
pyplot_feature_report.md                              this report
pyplot_feature_report.pdf                              PDF rendering
ngspice-46/src/frontend/com_pyplot.c                   (new)
ngspice-46/src/frontend/com_pyplot.h                   (new)
ngspice-46/src/frontend/plotting/pyplot.c              (new)
ngspice-46/src/frontend/plotting/pyplot.h              (new)
ngspice-46/src/frontend/commands.c                     (modified --- carries other commands too)
ngspice-46/src/frontend/plotting/plotit.c              (modified --- carries other plotting too)
ngspice-46/src/frontend/Makefile.am                    (modified)
ngspice-46/src/frontend/plotting/Makefile.am           (modified)
examples/pyplot_examples/                               the verifying example suite
```